OS-1

Process: চলমান Program.

↓

active entity → Process.

Process Management.

Program এবং **Process**

✓ HDD — RAM.

প্রতিটি Process:

NEW, READY, RUNNING, WAITING, TERMINATED

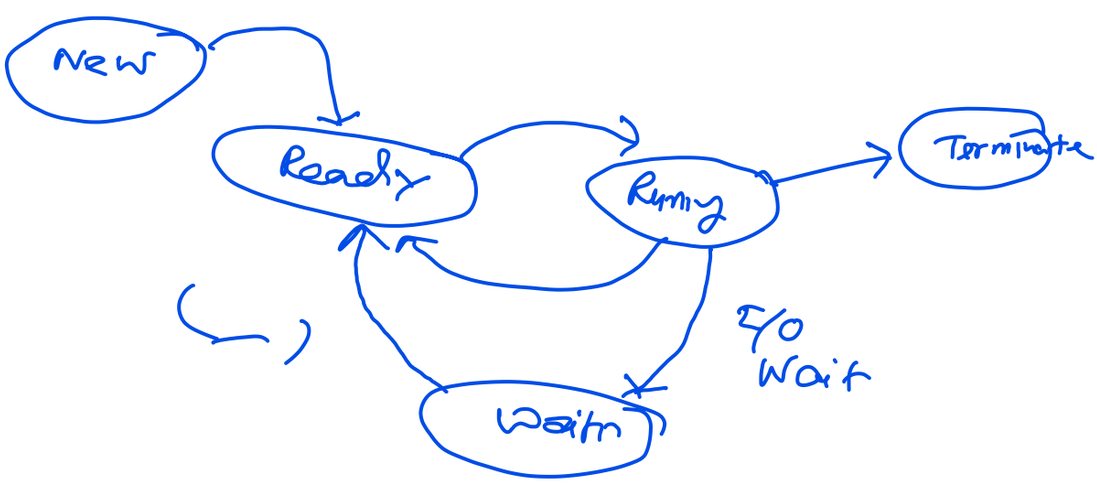
Process Table →

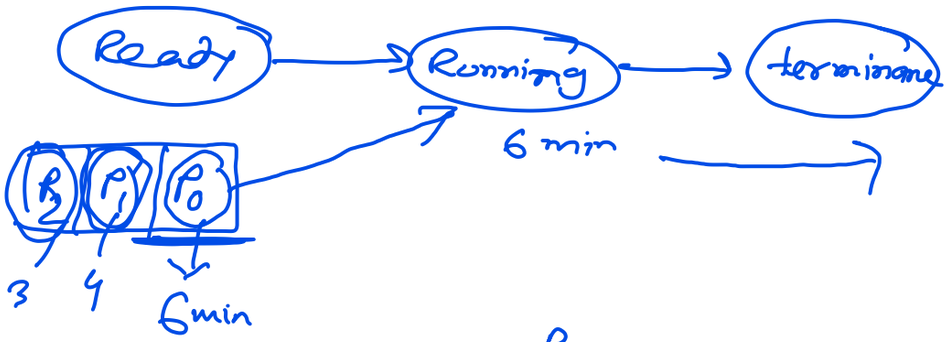
- Process.
- ✓ CPU, Memory
- ✓ Waiting.
- ✓ Running.
- ✓ Terminated

P_1 P_2 P_3 .... P_n

Scheduling Algorithm:

- → Non-Preemptive
- → Preemptive





CPU = 2 min

$P_0 = 2$
$P_1 = 2$
$P_2 = 2$

$P_1 = 2$
$P_0 = 4$
$P_2 = 1$

waiting

Arrival time:



P_0 = 6:58 → 7:00

arrival time = 7.00

6:40
6:50
6:55

Burst time: total time to execute a process on the CPU

P_0 = 4 min

Completion / Exit time:

Burst time = 40 min
waiting = 20 min

7:00 (arr) → 8:00 (Comp)

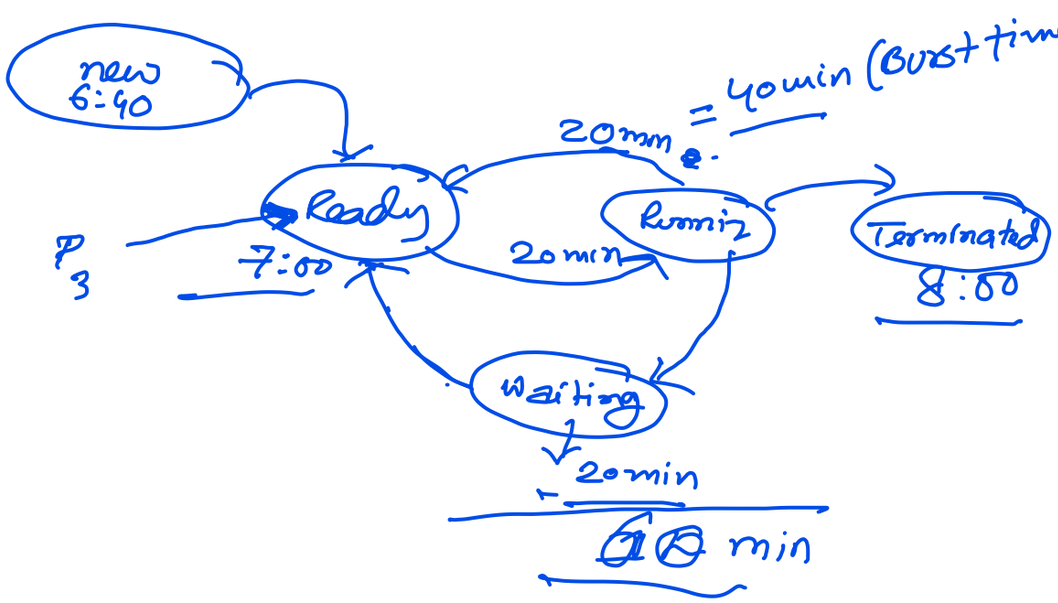
Turn around time:

= Exit time − Arrival time

= 8:00 − 7:00 = 1 h

⇒ Burst time + waiting = 40 + 20

= 60 m = 1 h



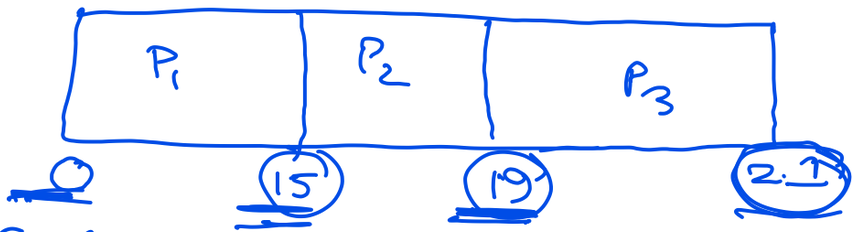
(1) FCFS (2) SJF (3) SRTF (4) Round Robin
(5) Priority.

(1) First-Come, First-Serve: (FCFS) : Non-Preemptive

Process	Burst Time (ms)
P_1	15
P_2	4
P_3	2.



Gantt chart:



Waiting time:

$P_1 = 0$

$P_2 = 15$

$P_3 = 19$

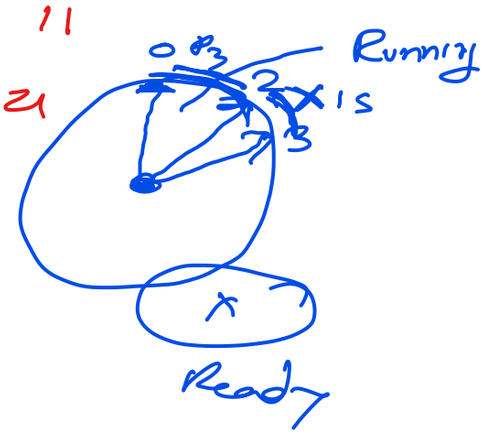
TAT, $P_1 = 15 + 0 = 15$

$P_2 = 4 + 15 = 19$

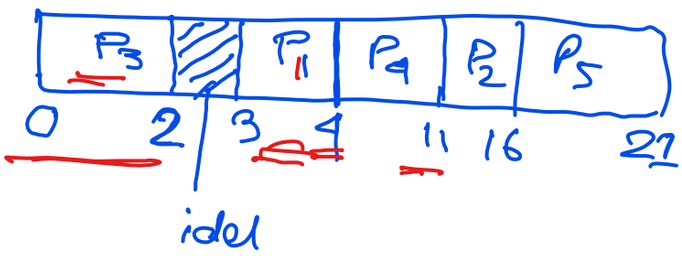
$P_3 = 2 + 19 = 21$

Ex2

Process	Arrival time	Busst time	Completion time	TAT
P_1	3	1	4	
P_2	4	5	16	
P_3	0	2	2	
P_4	3	7	11	
P_5	5	5	21	



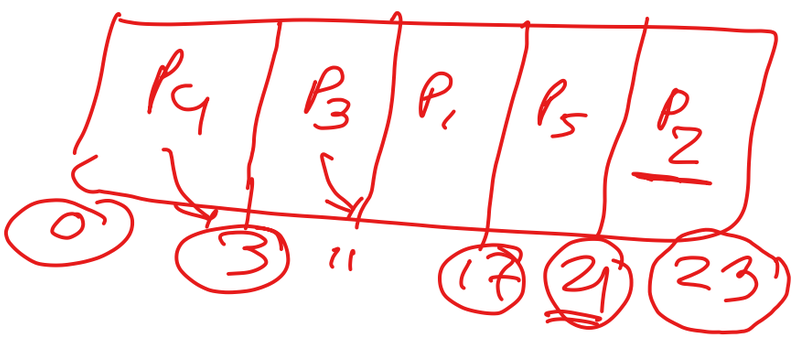
Grant chart



3 ↔ 4 = P_1, P_4

Process	Burst time	Arrival time
P_1	6	2
P_2	2	5
P_3	8	1
P_4	3	0
P_5	4	4

Grant chart:

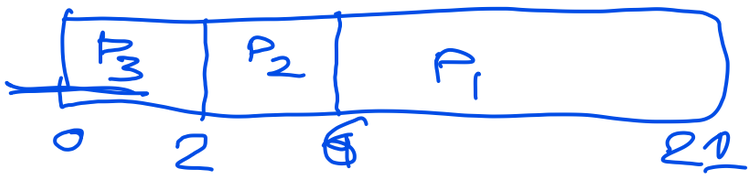


8:00

Comptl/Exit time = 23

Shortest-Job First scheduling: (Non-Preemptive)

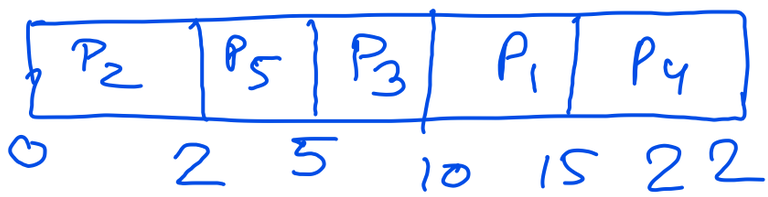
Process	Burst time
P_1	15
P_2	4
P_3	2



Ex2

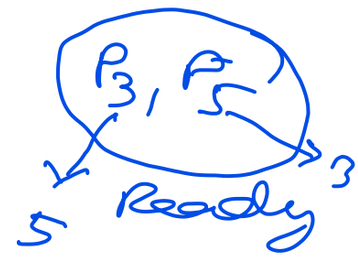
Process	arrival time	Burst time
P_1	4	5
P_2	0	2
P_3	1	5
P_4	6	7
P_5	2	3

Grant chart

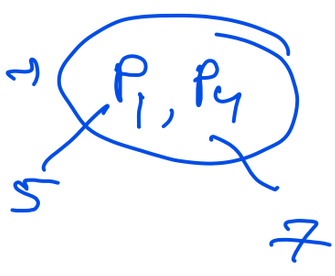




$0 \rightarrow 2$



$5 \rightarrow 10 \rightarrow$



$2 \rightarrow 5$

